Catastrophic cancellation

Catastrophic cancellation

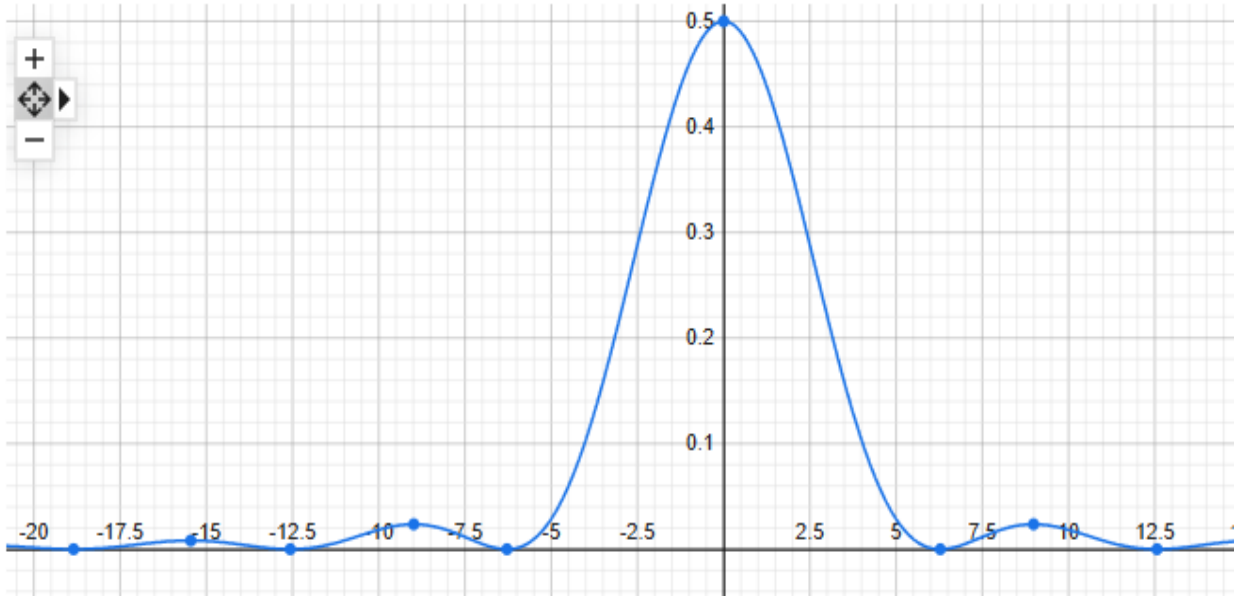
The Problem

When two numbers are very close in value, their difference is much smaller than either number individually. In floating-point arithmetic, this exposes the inherent imprecision in the representation, magnifying relative errors dramatically.

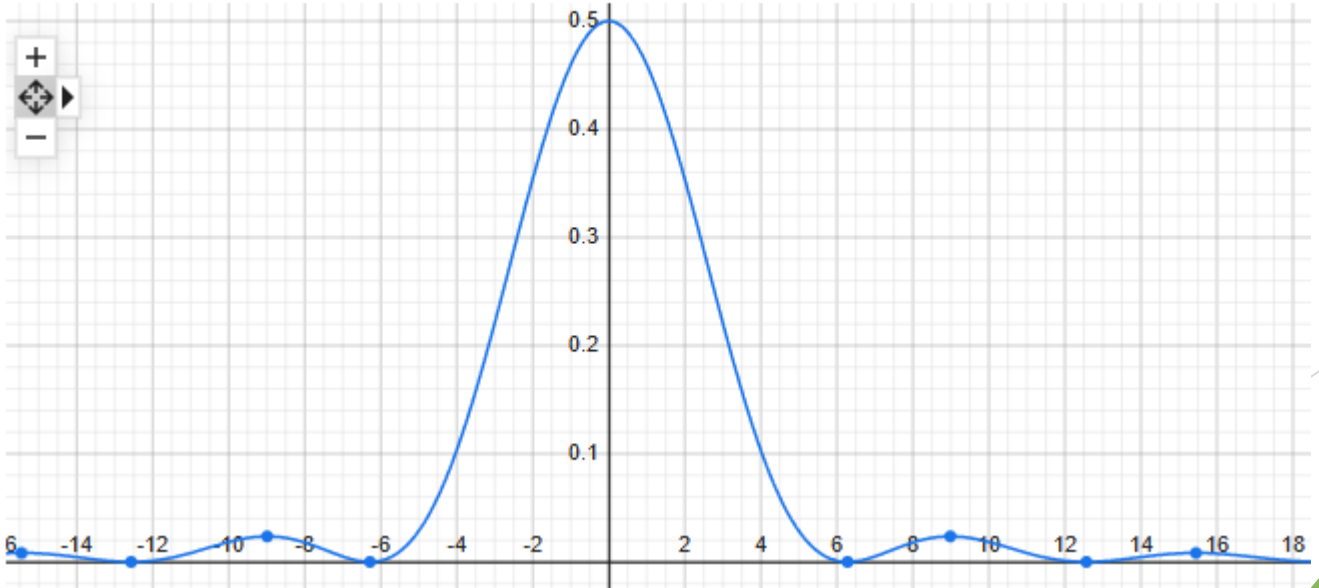
For very small step sizes h, subtractive forms like (f(x+h)-f(x))
suffer **catastrophic cancellation**

Consider: $f(x) = (1 - \cos(x)) / x^2$ for small values of x.

Graph for $(1 - \cos(x))/x^2$



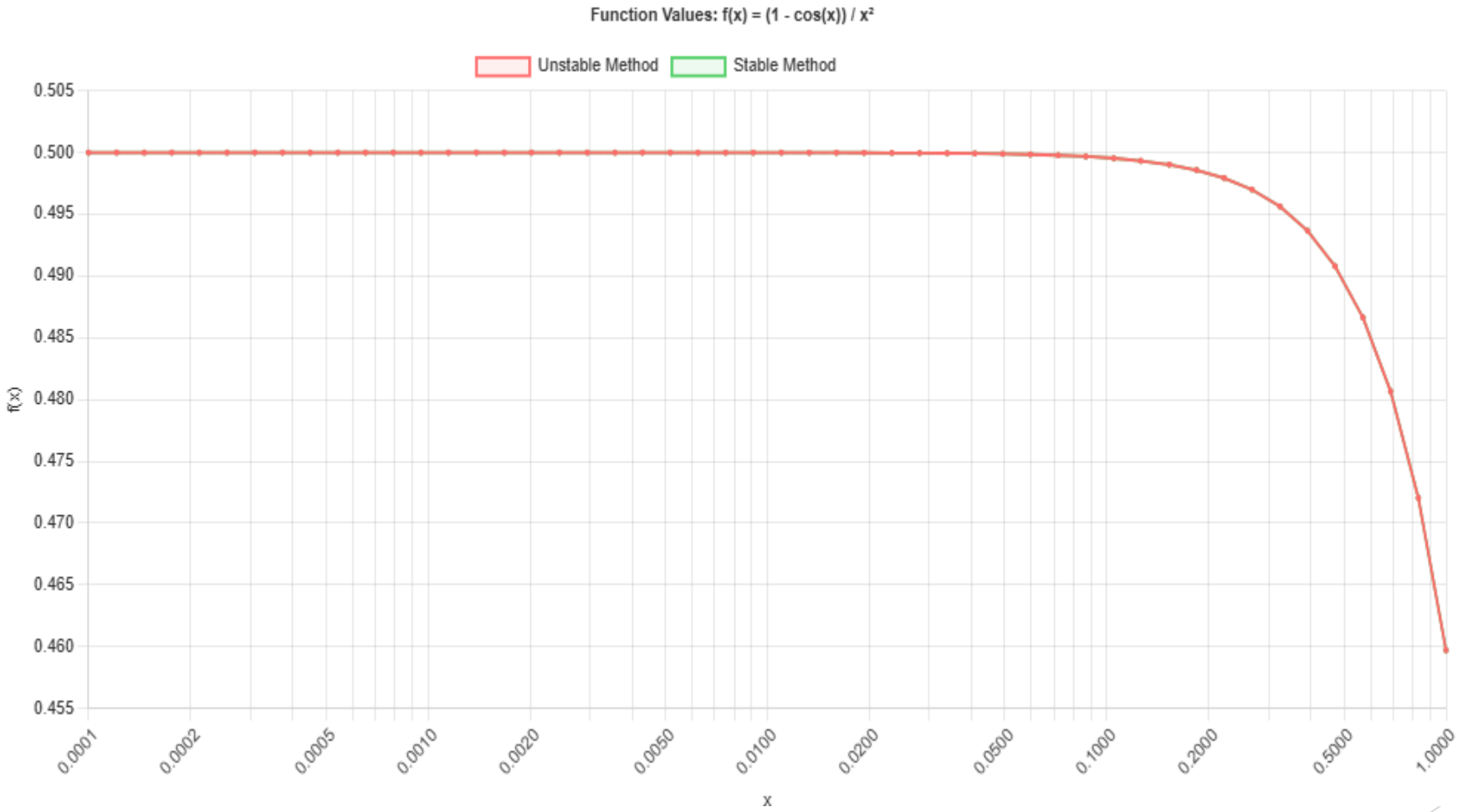
Graph for $2\sin^2(x/2)/x^2$



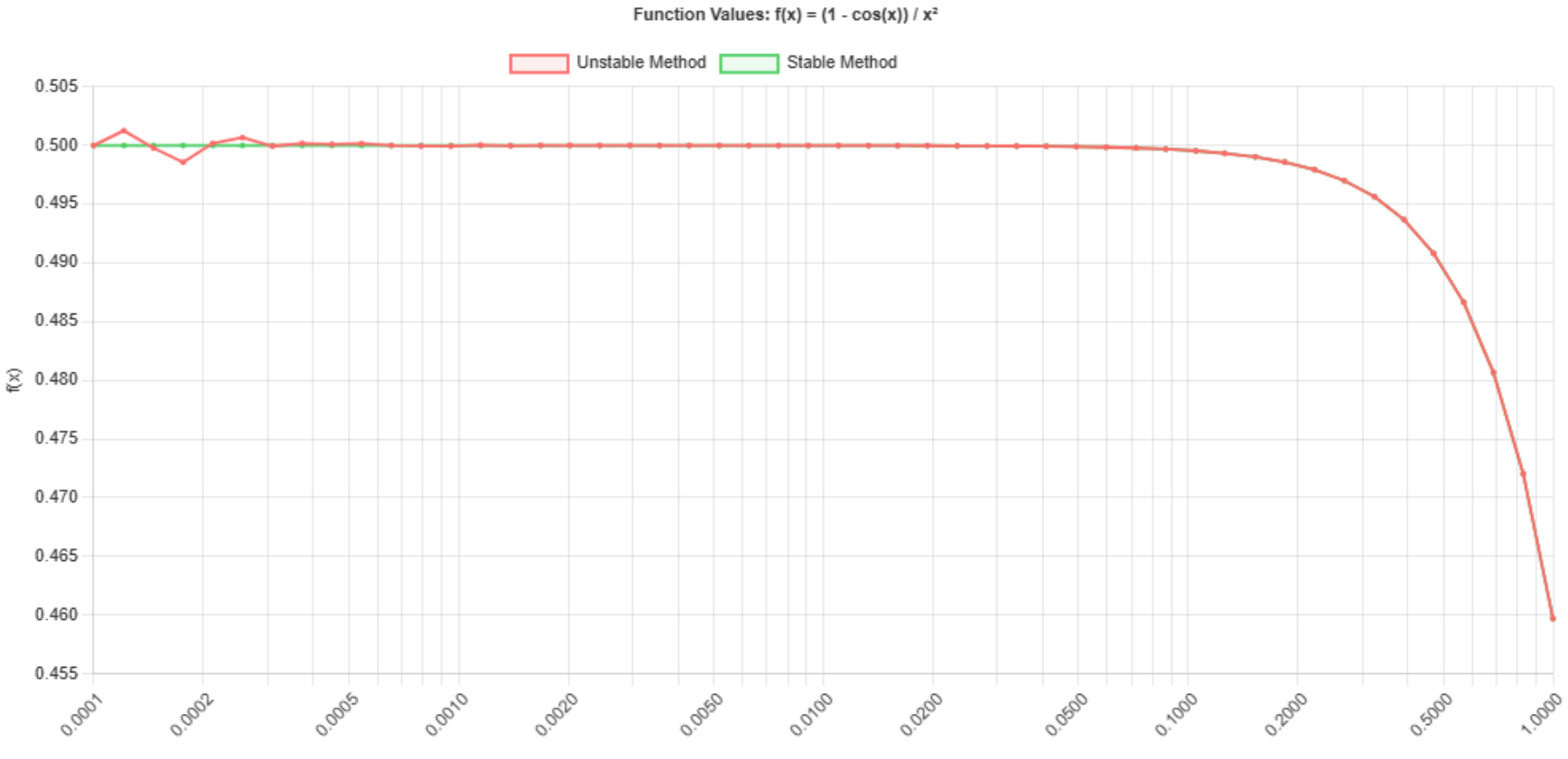
Calculation

f(x) = (1 - cos(x)) / x^2				x	cos(x)	x^2	1-cos(x)	(1 - cos(x)) / x^2	2*(SIN(x/2))^2/(x^2)
				1	0.540302306	1	0.459697694	0.459697694	0.459697694
				0.1	0.995004165	0.01	0.004995835	0.499583472	0.499583472
				0.01	0.99995	0.0001	4.99996E-05	0.499995833	0.499995833
				0.001	0.9999995	0.000001	5E-07	0.499999958	0.499999958
				0.0001	0.999999995	0.00000001	5E-09	0.499999997	0.5
				0.00001	1	1E-10	5E-11	0.500000041	0.5
				0.000001	1	1E-12	5.00044E-13	0.50004445	0.5
				0.0000001	1	1E-14	4.996E-15	0.499600361	0.5
				0.00000001	1	1E-16	0	0	0.5
				0.000000001	1	1E-18	0	0	0.5
				1E-10	1	1E-20	0	0	0.5
				1E-11	1	1E-22	0	0	0.5
				1E-12	1	1E-24	0	0	0.5
				1E-13	1	1E-26	0	0	0.5
				1E-14	1	1E-28	0	0	0.5
				1E-15	1	1E-30	0	0	0.5
				1E-16	1	1E-32	0	0	0.5
				1E-17	1	1E-34	0	0	0.5
				1E-18	1	1E-36	0	0	0.5
				1E-19	1	1E-38	0	0	0.5
				1E-20	1	1E-40	0	0	0.5

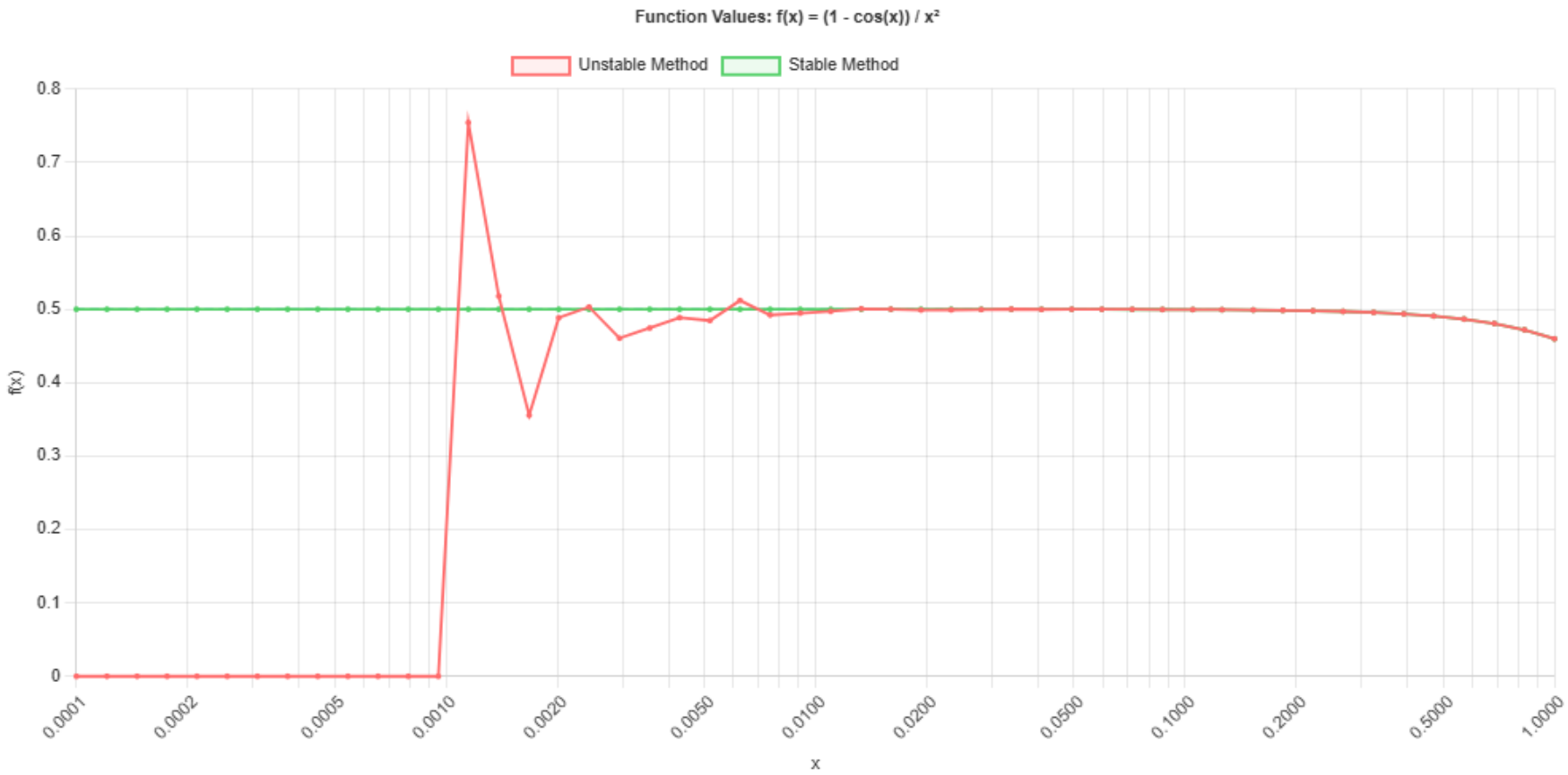
15 Digit Precision



10 Digit Precision



06 Digit Precision



Solutions

Mathematical Reformulation: Use algebraically equivalent expressions that avoid problematic subtractions

Higher Precision Arithmetic: Use double precision when needed

Series Expansions: Use Taylor series or other expansions as alternatives

Specialized Algorithms: Use numerically stable algorithms designed for specific problems